

Schema Anomalies

Esme Basnet

Credit and thank you go to Professor Bruce Irvin from Portland State University for generously providing me with this material.

Issues with Un-Normalized Data

Atomicity Anomaly

Update Anomaly

Insertion Anomaly

Deletion Anomaly

Atomicity Anomaly

Atomicity Anomaly - cells contain multiple sub-values packed together. If we ever need any of those sub-values then the DB will not help us (much) in finding and using them.

Atomicity Anomaly - Bank Customers

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ID	Name	Account IDs
1010	Joe Flint, Jane Flint	2590, 2013
5432	Jessica Bailey Leslie Marcus	992, 93292, 54

Atomicity Anomaly - Bank Customers

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ID	Name	Account IDs
1010	Joe Flint, Jane Flint	2590, 2013
5432	Jessica Bailey Leslie Marcus	992, 93292, 54

advice: avoid doing this by explicitly modeling the 1:n relationships both in the conceptual design and logical design.

Insertion Anomaly

Table is designed such that we cannot know/record some data until later

Insertion Anomaly - Superbowl Results

Table is designed such that we cannot know/record some data until later

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
55	Feb 7, 2021	LV	NULL	NULL	NULL	NULL	NULL	Raymond James Stadium	Tampa	Florida

Insertion Anomaly - Superbowl Results

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ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
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2	Jan 14 1968	II	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
55	Feb 7, 2021	LV	NULL	NULL	NULL	NULL	NULL	Raymond James Stadium	Tampa	Florida

advice: separate the “known later” dynamic data into a separate table.

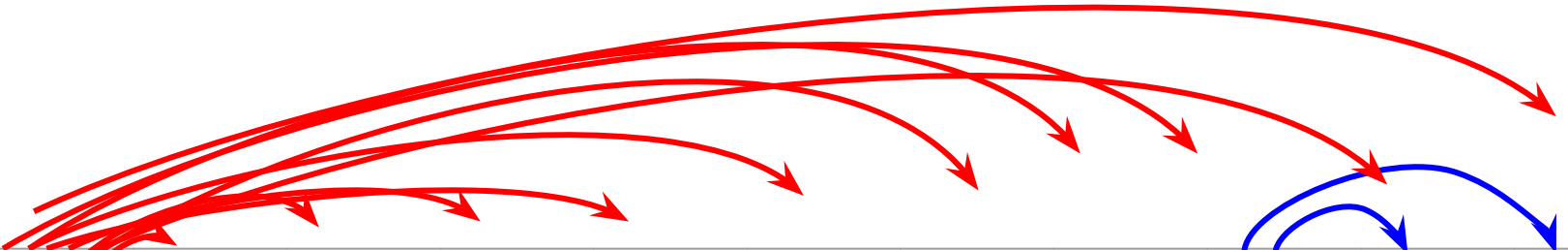


Update Anomaly

Concepts stored redundantly. So an update to one row might be missed in others

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
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Superbowls - Functional Dependencies



The diagram illustrates functional dependencies using red and blue arrows. Red arrows originate from the 'ID' column and point to 'Date', 'Title', 'Winner', 'score' (first instance), 'Loser', 'score' (second instance), 'MVP', 'Site', 'City', and 'State'. Blue arrows originate from the 'Site' column and point to 'City' and 'State'.

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

Superbowls - Troublesome Functional Dependencies

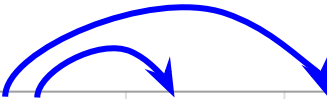


Diagram illustrating functional dependencies: Site → City and Site → State.

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

Update Anomaly

update to one row might be missed in others

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

advice: lift the troublesome functional dependency data into a separate table. In this case, create a “stadium” table.

Deletion Anomaly

If we delete a row, then we also delete other concepts (e.g., Los Angeles)

ID	Date	Title	Winner	score	Loser	score	MVP	Site	City	State
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	Memorial Coliseum	Los Angeles	California
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	Miami Orange Bowl	Miami	Florida
3	Jan 12 1969	III (3)	New York Jets	16	Baltimore Colts	7	Joe Namath+	Orange Bowl	Miami	Florida

advice: model important concepts with their own table(s). “stadium”, “city”, “state”, “player”, etc.

Normalization of Superbowl Data

Superbowls

<u>ID</u>	Date	Title	Winner	Score	Loser	Score	MVP	Site_ID
1	Jan 15 1967	I (1)	Green Bay Packers	35	Kansas City Chiefs	10	Bart Starr+	1
2	Jan 14 1968	II (2)	Green Bay Packers	33	Oakland Raiders	14	Bart Starr+	2
...	...	...	...					

Stadiums

<u>ID</u>	Name	City	State
1	Memorial Coliseum	Los Angeles	CA
2	Miami Orange Bowl	Miami	FL
...	...	...	...

Note: another “troublesome” functional dependency

Stadiums



<u>ID</u>	Name	City	State
1	Memorial Coliseum	Los Angeles	CA
2	Miami Orange Bowl	Miami	FL
...	...	...	...

Un-normalized data is very common!

E.g.: data scientists often work with single-table data sets

Does that mean that it is a good thing? Or at least not a bad thing?

...depends on what you are trying to accomplish!

What are your priorities?

Simplicity? (fewer tables, better read performance)

Efficiency? (eliminate redundancies, better write performance)

Are Updates, Insertions and Deletions really a concern for your system?

Summary: Issues with un-Normalized Data

Atomicity Anomaly - cells contain multiple sub-values packed together

Update Anomaly - update to a row needs might fail to happen to all affected rows

Insertion Anomaly - we only record some facts when an entire row is inserted

Deletion Anomaly - deleting a row might delete knowledge of other entities

De-normalization: making a database less-normalized to increase read performance at the expense of redundancy and write performance.

